HARAM NAJAM KHAN

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SUMMARY:

I am a motivated Computer Engineering student with strong analytical and problem-solving skills, and experience in projects involving machine learning and optimization techniques. Known for delivering quality work and effective presentations. Eager to apply technical knowledge and continuously learn in a challenging professional environment.

WORK EXPERIENCE:

NRTC Haripur– *Intern* *Nov 2025*

- Interned at NRTC, gaining hands-on experience in LabVIEW, signal processing, and communication systems through practical lab work and system analysis.

EDUCATION:

- **COMSATS University Islamabad, Abbottabad Campus**

Bachelor of Science in Computer Engineering – **Present**

- **ARMY PUBLIC SCHOOL AND COLLEGE, PMA KAKUL ABBOTTABAD (2022)**

FSc in Pre Engineering – **Grade A**

PROJECTS

Multi-Band GPS Signal Modelling and Transmission via Software Defined Radio (L1, L2, L5) — Final Year Project (in collaboration with NRTC) *Sep 2025 – Present*

- Designing and implementing multi-band GPS signal models (L1, L2, L5) using **Software Defined Radio (SDR)** and **LabVIEW**
- Simulating and analysing **GPS signal generation, modulation, and transmission** in a controlled lab environment.
- Applying signal processing techniques for **accurate modelling and real-time transmission**.
- Collaborating with **NRTC engineers** to align project implementation with industry practices.

Current Mirror Design and Analysis — VLSI Design Project *Nov 2025*

- Designed and simulated current mirror circuits using **Cadence tools**.
- Analysed circuit performance in terms of **current stability, scaling, and efficiency.**
- Gained practical exposure to **analog IC design concepts and VLSI workflows.**

Automatic Parking System — Project Planning & Management *Nov 2025*

- Developed a project plan for an **automated parking solution** to optimize space utilization.
- Defined system requirements, workflow, and implementation strategy.
- Applied project management concepts including **planning, scheduling, and resource allocation.**

Fibonacci Series Calculation using PIC18 Microcontroller — Microprocessor System Project *June 2025*

- Implemented Fibonacci sequence generation using **PIC18 microcontroller**
- **Programmed in assembly** to perform iterative calculations
- Simulated and tested output for accuracy and efficiency

Disease Risk Prediction System — Machine Learning Project *June 2025*

- Developed predictive models for diabetes, heart disease, and cancer risk assessment using neural networks
- Trained and evaluated models on healthcare datasets using appropriate preprocessing techniques
- Assessed model performance using metrics such as accuracy, precision, recall, and F1-score

Traffic Signal Timing Optimization using CSP and Simulated Annealing —Artificial Intelligence Project Oct-Nov *2024*

- Modelled the traffic signal control problem using **Constraint Satisfaction Problem (CSP)** techniques
- Implemented **Simulated Annealing** to optimize signal timing for improved traffic flow.
- Reduced congestion by finding near-optimal solutions under real-world constraints.
- Analysed system performance and efficiency through simulation results.

Car Buying & Selling Website — Database Systems Project *Nov 2024*

- Designed and developed a **web-based platform** for buying and selling cars.
- Implemented database structure for **user data, listings, and transactions**.
- Integrated frontend and backend for a **dynamic user experience.**

LEADERSHIP/EXTRACURRICULAR

- Serving as **Chairperson**, IEEE Student Branch, COMSATS Abbottabad *Nov 2025- P r e s e n t*
- Served as **Chairperson**, IEEE -Women In Engineering (WIE), COMSATS Abbottabad *July 2025- Nov 2025*
- **Head Student Editor, Computer Engineering Department Newsletter**, led the editorial team and managed content creation and publication. Jan 2026
- Organized **Innovate Expo and Open House 2025** in collaboration with CE Department. July 2025

SKILLS

- **Programming Languages:** C, C++, Python, Assembly, HTML, CSS, SQL, Matlab.
- **Libraries:** PyTorch, Matplotlib, Keras, NumPy, Pandas.
- **Hardware Platforms:** Arduino, Pic Micro-controllers (PIC18), Raspberry pi, Software Defined Radio USRP.
- **Software and Tools:** MATLAB, Google Colab, Jupyter Notebook, Visual Studio, Adobe Illustrator, Linux, Cadence, Proteus, Canva, LabVIEW, MPLAB X IDE, Wireshark, Packet Tracer, GitHub.
- **Soft Skills:** Leadership, Communication, Teamwork, Problem Solving, Critical Thinking, Time Management, Presentation, Public Speaking.

CERTIFICATIONS:

- Mastering in AutoCAD-4-day workshop **(Organizer).**
- Design to Manufacture Full Cycle Development of State-of-the-Art Embedded Projects **(Organizer).**
- Workshop Unleashing Creativity with Generative AI.
- Modern trends In Career Development Session **(Conducted)**